Robust Online Predictive Visual Servoing for Autonomous Landing of a Rotor UAV

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***Abstract*—This paper presents a novel approach that integrates image-based visual servoing (IBVS) with online model predictive control (MPC) for landing a rotor Unmanned Aerial Vehicle (UAV) on maritime vessels, which ensures long-term visibility of the landing tag and enhances robustness against external disturbance. In contrast to existing studies that overlook the influence of UAV attitude changes on target visibility, this work distinguishes itself by comprehensively considering the attitude changes and the visibility constraints in the optimization framework, thereby effectively averting target loss. Furthermore, this paper contributes to the field by analyzing the input-to-state stability of the proposed approach under bounded disturbances, where the upper bound of disturbances is provided when the convergence domain is specified. To address the specific challenge of landing from high altitudes, a two-stage control strategy leveraging a self-designed nested landing tag is introduced. This strategy enables fast and accurate landings with smooth velocity transitions. Additionally, a soft constraint method is introduced to prioritize solution feasibility in online optimization. Extensive numerical simulations, software-in-the-loop (SITL) simulations, and field experiments validate the feasibility and superiority of our proposed method, underscoring its potential for practical implementation.**

***Index Terms*—Image-based visual servoing (IBVS), model predictive control (MPC), image moments, bounded disturbance, robust control.**

I. Introduction

THE autonomous flight of UAVs has attracted extensive attention, and the autonomous landing is an important prerequisite for the realization of the autonomy. Actually, the landing phase is the most accident-prone stage for the UAVs. Therefore, it is necessary to study the accurate autonomous landing of UAVs. Many works have employed global positioning system (GPS) to calculate relative positions for autonomous landing [1], [2]. However, there usually exist large errors during the positioning, and the signal interruption is also the crux. Alternatively, the VICON motion capture system is adopted to provide accurate positioning information for UAV landing in [3], but the system is not suitable for outdoor experiments. Besides, the radar can guide the landing of UAVs [4], but it is susceptible to electromagnetic interference and its widespread adoption is hindered by its high cost. In comparison, vision sensors offer a more cost-effective solution by enabling non-contact measurements, thus enhancing the reliability of autonomous landing.

With the aid of vision sensors, the visual servoing scheme can be adopted to design the corresponding controller. It is a method based on the visual feedback, composed of the visual signal processing and robot control. According to the difference in the feedback information, the visual servoing methods can be mainly divided into position-based visual servoing (PBVS) and image-based visual servoing (IBVS) [5]. The PBVS method defines the visual error in the 3-D Cartesian space, which has been extensively applied to the control of robotic manipulators [6], [7], [8] and UAVs [9], [10], [11]. However, the errors caused by the camera calibration are inevitable, which result in positioning bias and will affect the landing accuracy ultimately. In contrast, the IBVS method directly design the controller in the 2-D image plane, which is robust to the camera calibration errors. Similarly, this method has also been widely used in robotic manipulators [12], [13], [14] and UAVs [15], [16], [17].

Based on the advantages of the IBVS method, there is a vast literature on the use of it for autonomous landing of UAVs, including the landing on static platforms [18], [19], [20] and moving platforms [21], [22], [23]. However, the classical IBVS method does not take the model dynamic constraints and visibility constraints into consideration, which may cause the target to be out of sight. To solve the problem, MPC is introduced to handle these constraints, thereby improving the performance of IBVS. Roque et al. [24] employ linear MPC to address under-actuation issues in UAVs when combining the IBVS modules, but their focus does not encompass visibility constraints. Furthermore, Sheng et al. [25] propose an observer-based MPC scheme for the IBVS of a quadrotor, which improves the visibility by bounding the roll and pitch angles. However, it lacks strict theoretical guarantee. Besides, Li et al. [26] adopt model predictive IBVS for a quadrotor UAV equipped with a fixed camera, and Zhang et al. [27] propose a robust nonlinear MPC scheme for the visual servoing of quadrotors subject to external disturbances. Both of them take the visibility constraints into the MPC optimization problems and the visibility

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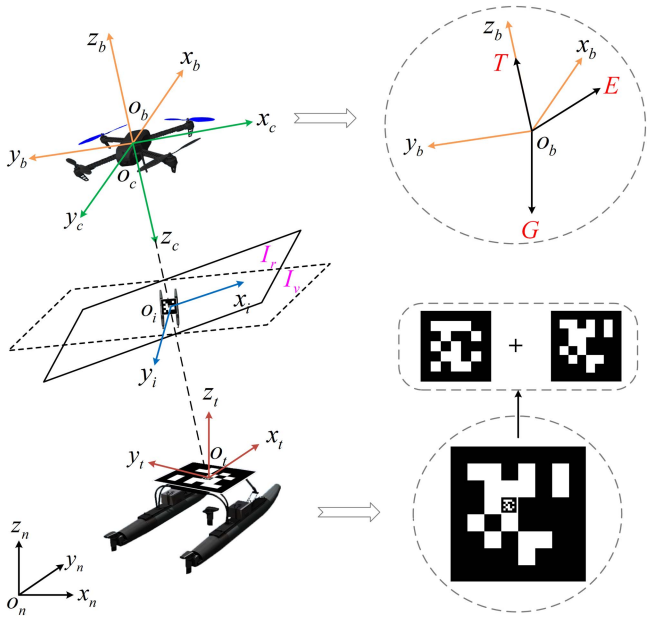


Fig. 1. Autonomous landing of a rotor UAV on a boat.

is theoretically ensured. However, they ignore the impact of pose changes and directly consider the visibility constraints in the virtual image plane, which does not represent the true field of view (FOV) of the camera. Particularly, as the target approaches the image edge, the UAV attitude changes may cause the target deviating from the actual image plane. Consequently, the target will be out of sight and thereby disrupting control continuity.

Actually, accurately landing a rotor UAV poses several challenges that need to be addressed. Firstly, the coupling of rotation with translation in the rotor UAV affects the camera attitude due to the UAV's motion, complicating visibility constraints analysis. Existing approaches often simplify this by considering them directly in the virtual image plane, but this cannot guarantee true visibility. Additionally, external factors like wind disturbances introduce inaccuracies in the kinematic model, making it crucial to ensure landing accuracy under disturbances. Furthermore, landing from high altitudes not only requires the guarantee of target recognition at different altitudes, but also demands fast and precise landing capabilities.

To this end, we propose a robust online MPC-IBVS approach for the landing of a rotor UAV in this paper, which guarantees long-term visibility of the landing tag while being robust to the external disturbances. In addition, two different controllers are designed for the problem of landing from a high altitude, where the first one aims at fast landing while the second one is for accurate landing, and the velocity can be smoothly transitioned when switching the first one to the second one. Through comparative experiments based on the numerical simulation, we show that our method can better keep the target within the image range when considering the coupled property of the rotor UAV, and can make the final state converge to a small terminal region under the condition of bounded disturbances. Last but not least, both the SITL simulations and field experiments are conducted to further verify the feasibility of our method. The contributions of this paper are summarized as follows:

- In this paper, we propose a novel robust MPC-IBVS approach that ensures long-term visibility throughout the entire landing process. In contrast to [27] that directly analyzes the visibility constraints in the virtual image plane, our method incorporates considerations for the impact of UAV attitude changes. This enables the attainment of stringent theoretical guarantees regarding target visibility and thereby preventing control disruption caused by target loss.
- Furthermore, we analyze the input-to-state stability of the proposed approach under bounded disturbance. We also provide the upper bound of the disturbance when specifying the final convergence domain, thereby empowering the rotor UAV to achieve accurate landing.
- To address the challenge of landing from a high altitude, we present a two-stage control strategy based on a self-designed nested landing tag, which facilitates fast and accurate landing. Additionally, we introduce a soft constraint method to prioritize the solution feasibility, effectively addressing online optimization considerations.

The rest of this paper is organized as follows. Section II introduces the dynamic model of the rotor UAV and the image kinematics. In Section III, the details for the design of MPC-IBVS controller is described, simultaneously, the feasibility and stability of the controller are analyzed as well. Sections IV, V, and VI present the results of numerical simulation, SITL simulation and field experiment, respectively. Finally, Section VII concludes this paper.

II. Problem Formulation

In this work, a method integrating MPC and IBVS is proposed to achieve the autonomous landing of a quadrotor UAV equipped with a downward camera on a boat. As Fig. 1 shows, $\mathcal{F}_n = \{o_n, x_n, y_n, z_n\}$ is the inertial frame, the body frame $\mathcal{F}_b = \{o_b, x_b, y_b, z_b\}$ is fixed on the UAV body, where o_b is located at the center of gravity of the rotor UAV and x_b is along the head direction of the UAV. The origin o_c of the camera frame $\mathcal{F}_c = \{o_c, x_c, y_c, z_c\}$ is located at the optical center of the camera, which is assumed to be coincide with o_b since the camera is attached underneath the geometric center of the UAV. Besides, the origin o_i of the image frame $\mathcal{F}_i = \{o_i, x_i, y_i\}$ lies at the image center, and the origin o_t of the tag frame $\mathcal{F}_t = \{o_t, x_t, y_t, z_t\}$ lies at the geometric center of the tag. Besides, I_r is the real image plane and I_v represents the virtual image plane that is horizontal. In order to achieve the landing from a high altitude, a nested tag is designed to ensure the detection within a large altitude range. This tag is a combination of two different types of AprilTags [28] whose geometric centers are coincident, the big one belongs to family 36h11 while the other belongs to family 25h9.

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A. Dynamic Model of Rotor UAV

The dynamic model of the rotor UAV used in this work has been widely used in [27], [29], which is

$$\begin{cases} \dot{P} = V \\ m\dot{V} = T \cdot R \cdot I_3 - mg \cdot I_3 \\ \dot{R} = R \cdot [\Omega^b]_\times \\ J \cdot \dot{\Omega}^b = -[\Omega^b]_\times \cdot J \cdot \Omega^b + \tau \end{cases}, \tag{1}$$

where $P \in \mathbb{R}^3$ and $V \in \mathbb{R}^3$ represent the position and velocity vector of rotor UAV in $\mathcal{F}_n$, respectively. T and $\tau \in \mathbb{R}^3$ are the net thrust force and the torque inputs, respectively. Besides, m and $J \in \mathbb{R}^{3\times3}$ are the mass and moment of inertia for the rotor UAV, g is gravity acceleration, R is rotation matrix from $\mathcal{F}_b$ to $\mathcal{F}_n$, $I_3 = [0, 0, 1]^\top$, and $\Omega^b \in \mathbb{R}^3$ is the angular velocity in $\mathcal{F}_b$, $[\cdot]_\times$ represents skew symmetric matrix.

Besides, the external disturbance E is considered in the model, and the relationship between T, E and gravity G is shown as Fig. 1. Denote the yaw, pitch and roll angles of the UAV by $\{\psi, \theta, \phi\}$ in turn. The acceleration of UAV and external disturbance in the j-axis of $\mathcal{F}_b$ are denoted by a_j and E_j, respectively. Then according to the second equation of (1), there exists

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \frac{T}{m} \begin{bmatrix} \cos\phi \cdot \sin\theta \\ -\sin\phi \\ \cos\phi \cdot \cos\theta \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} + \frac{1}{m} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}. \tag{2}$$

Based on (2), the relationship between acceleration and attitude can be obtained, which is used in Section III-A. Note that this work focuses on the design of outer-loop velocity controller involving $V^b = [V_x^b, V_y^b, V_z^b, \omega_z^b]$, which is applicable to different types of rotor UAVs, and the inner-loop control is implemented by the commonly used autopilot system such as Pixhawk.[1]

B. Image Kinematics

IBVS is a method which maps the time variation of the feature point in the image into the control input directly. Since the feature point in $\mathcal{F}_i$, denoted by $s_1 = [u_1, v_1]^\top$, is affected by the attitude change of the rotor UAV. Therefore, the virtual camera method [30] is introduced to solve the problem, and the corresponding feature point in the virtual image plane, denoted by $s_2 = [u_2, v_2]^\top$, can be obtained.

Furthermore, denote the linear velocity vector and angular velocity vector of the UAV in $\mathcal{F}_c^v$ by $V^c = [V_x^c, V_y^c, V_z^c]^\top$ and $\Omega^c = [\omega_x^c, \omega_y^c, \omega_z^c]^\top$, respectively. Then the following IBVS model can be obtained:

$$\dot{s} = L_s(u_2, v_2) \cdot \begin{bmatrix} V^c \\ \Omega^c \end{bmatrix}, \tag{3}$$

where $L_s(u_2, v_2) \in \mathbb{R}^{2\times6}$ is the image Jacobian matrix.

Next, four corner points of the landing tag are extracted to be the feature points, which are denoted by $[u_{2l}, v_{2l}]^\top, l = \{1, 2, 3, 4\}$, in the virtual image plane. After that, the centroid coordinates of the tag can be represented as $[u_c, v_c]^\top = [\frac{1}{4}\sum_{l=1}^4 u_{2l}, \frac{1}{4}\sum_{l=1}^4 v_{2l}]^\top$.

Since the image feature is insensitive to the change in altitude, the convergence rate will be low in altitude [31]. Therefore, the image moments are introduced to address the problem, they are defined as follows:

$$q_h = \sqrt{\frac{a^*}{a}}, \quad q_u = \frac{u_c}{f} q_h, \quad q_v = \frac{v_c}{f} q_h, \tag{4}$$

where $a = \mu_{20} + \mu_{02}$, a^* is the desired constant value of a corresponding to a given desired altitude z^*, f represents the focal length of the camera and $\mu_{ij} = \sum_{l=1}^4 (u_{2l} - u_c)^i (v_{2l} - v_c)^j$.

Since the four selected image points are in a coplanar square, then the moment feature for yaw motion defined in [31], which is $q_\psi = \frac{1}{2}\arctan\left(\frac{2\mu_{11}}{\mu_{20}-\mu_{02}}\right)$, will not uniquely determine the heading direction of the UAV when $q_\psi = 0$. Therefore, we redefine it as $q_\psi = a\tan 2(v_{21} + v_{22} - 2v_c, u_{21} + u_{22} - 2u_c)$.

Based on the above definitions, the image moment vector can be expressed as

$$q = [q_u, q_v, q_h, q_\psi]^\top, \tag{5}$$

and the IBVS model can be rewritten as

$$\dot{q} = \begin{bmatrix} \frac{1}{z^*} V_y^b - q_v \cdot \omega_z^b \\ \frac{1}{z^*} V_x^b + q_u \cdot \omega_z^b \\ \frac{1}{z^*} V_z^b \\ \omega_z^b \end{bmatrix} \tag{6}$$

where $V^b = [V_x^b, V_y^b, V_z^b]$ and ω_z^b are control inputs defined in $\mathcal{F}_b$, z^* is the desired height of the rotor UAV relative to the landing tag in $\mathcal{F}_c$. In this work, it corresponds to the height at which the small tag is just detected during the landing.

C. MPC-IBVS Framework

The framework integrating MPC and IBVS is shown in Fig. 2. Utilizing a downward camera mounted on a rotor UAV enables the capture of a boat with a landing tag. The AprilTags detection algorithm is employed to extract the feature point s_1 of the tag, and through the rotation of θ and ϕ, the virtual point s_2 can be determined. Subsequently, the image moments are calculated based on (4), leading to the establishment of the IBVS model described by (6).

Based on the existing MPC-IBVS framework [24], both attitude changes and external disturbances are considered in our framework for long-term visibility and robust landing control. Typically, the existing MPC-IBVS framework mainly consists of four modules: The cost function emphasizes the convergence of image moments and the control input variances. The IBVS model is leveraged to predict the future state of the feature point for receding horizon optimization. The dynamic constraints are incorporated to prevent significant attitude adjustments from causing the UAV to roll over. The visibility constraints are considered due to the limited FOV of the downward camera. Furthermore, two additional modules have been integrated into this work, highlighted in blue. Unlike prior studies that directly analyze the visibility constraints in the virtual image plane,

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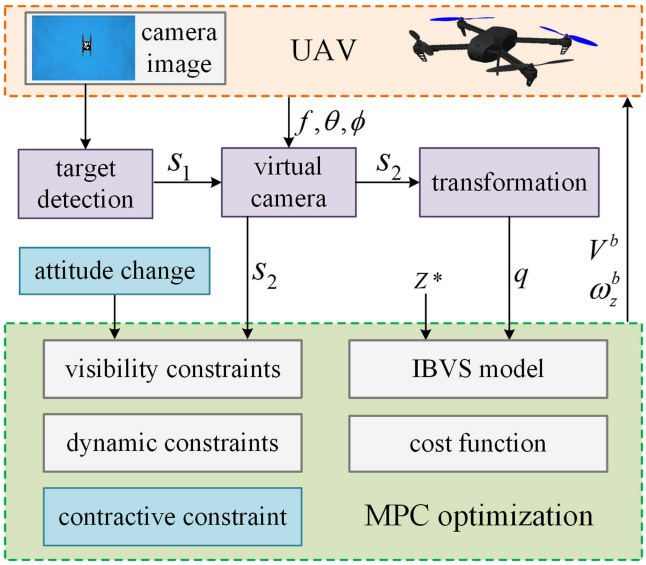


Fig. 2. Proposed MPC-IBVS framework.

this work accounts for the impact of UAV attitude changes on the visibility constraints, providing a true visibility guarantee. To handle external disturbances, a contractive constraint has been devised to achieve robust landing control, as detailed in Section III. Finally, the optimized control inputs, V^b and ω_z^b in (6), are used for the landing control of the rotor UAV.

III. MPC-IBVS Controller

A. MPC Problem Formulation

Denote the sampling time by dt and control vector by $U^\top = [V_x^b, V_y^b, V_z^b, \omega_z^b]$. According to (6), the discretized nominal system in time-step $k+1$ can be given as:

$$q_{k+1} = q_k + dt \cdot \dot{q}_k = \mathrm{f}_q(q_k, U_k). \tag{7}$$

Furthermore, the perturbed system is denoted by

$$q_{k+1} = \mathrm{f}_q(q_k, U_k) + \xi_k, \tag{8}$$

where ξ_k belongs to a bounded set $\Xi = \{\xi \in \mathbb{R}^{4\times 1} | \, \|\xi_k\| \leq \bar{\xi}\}$, and $\bar{\xi}$ is a positive scalar.

For convenience of understanding the meaning of ξ_k, we denote the real pose and measured pose of the UAV by $\{x_k, y_k, z_k, \psi_k\}$ and $\{\hat{x}_k, \hat{y}_k, \hat{z}_k, \hat{\psi}_k\}$, respectively. Besides, the error is defined as $\{\Delta x_k, \Delta y_k, \Delta z_k, \Delta\psi_k\}$. Actually, there also exists $q_{h,k} = \frac{z_k}{z^*}$ in addition to (4). Then the error of $q_{h,k}$ can be represented as

$$\Delta q_{h,k} = \frac{\Delta z_k}{z^*} \tag{9}$$

According to the perspective projection, the error of u_{ck} can be written as:

$$\Delta u_{c,k} = f\left(\frac{x_k}{z_k} - \frac{\hat{x}_k}{\hat{z}_k}\right) = f\left(\frac{\Delta x_k}{z_k} - \hat{x}_k \frac{\Delta z_k}{z_k \cdot \hat{z}_k}\right) \tag{10}$$

After that, the error of q_u can be obtained:

$$\Delta q_{u,k} = \tfrac{1}{f}\left(q_{h,k} \cdot u_{c,k} - \hat{q}_{h,k} \cdot \hat{u}_{c,k}\right) = \tfrac{1}{f}\tfrac{z_k}{z^*}\Delta u_{c,k} + \tfrac{\hat{x}_k}{\hat{z}_k}\Delta q_{h,k}. \tag{11}$$

Similarly, the error of q_v can be represented as

$$\Delta q_{v,k} = \frac{1}{f}\frac{z_k}{z^*}\Delta v_{c,k} + \frac{\hat{y}_k}{\hat{z}_k}\Delta q_{h,k}. \tag{12}$$

Besides, there exists $q_{\psi,k} = \Delta\psi_k$. To sum up, $\|\xi_k\|$ can be given as follows:

$$\|\xi_k\| = \sqrt{\Delta q_{h,k}^2 + \Delta q_{u,k}^2 + \Delta q_{v,k}^2 + \Delta q_{\psi,k}^2}. \tag{13}$$

In order to achieve the accurate landing of the UAV, q is required to converge to $q_d = [0, 0, 1, 0]^\top$. Then the error is defined as

$$\tilde{q}_{k+i|k} = q_d - \hat{q}_{k+i|k}, \tag{14}$$

where the predicted state $\hat{q}_{k+i|k}$ can be obtained based on the nominal system:

$$\hat{q}_{k+i|k} = \mathrm{f}_q(\hat{q}_{k+i-1|k}, U_{k+i-1|k}). \tag{15}$$

Now, the cost function can be defined as

$$J(\tilde{q}_{k|k}, \mathbf{U}_k) = \sum_{i=1}^{Np} J_r(\tilde{q}_{k+i|k}, U_{k+i-1|k}), \tag{16}$$

where

$$J_r(\tilde{q}_{k+i|k}, U_{k+i-1|k}) = \|\tilde{q}_{k+i|k}\|_Q^2 + \|U_{k+i-1|k}\|_R^2, \tag{17}$$

the control sequence is

$$\mathbf{U}_k = \{U_{k|k}, U_{k+1|k}, \ldots, U_{k+N_p-1|k}\}, \tag{18}$$

N_p is the prediction horizon, $Q, R \in \mathbb{R}^{4\times 4}$ are positive definite diagonal matrices, $\|\tilde{q}\|_Q^2 = \tilde{q}^\top Q \tilde{q}$ and similar to $\|U\|_R^2$. Note that $J = 0$ if and only if $\tilde{q}_{k+i|k} = 0$ and $U_{k+i-1|k} = 0$ for $i = 1, \ldots, N_p$. That is, the control input of the UAV is 0 when it reaches desired state.

After that, the constraints are taken into consideration. Denoting the control constraint set by U_{set}, then we have

$$U_k \in U_{set} \subseteq \mathbb{R}^{4\times 1}. \tag{19}$$

To prevent the UAV from ascending, we set the maximum value of V_z^b to 0. Furthermore, denote the acceleration constraint set by a_{set}, where

$$a_k \in a_{set} \subseteq \mathbb{R}^{4\times 1}. \tag{20}$$

Assuming uniform acceleration during the sampling time, we have

$$a_{k+1} = \frac{U_{k+1} - U_k}{dt}. \tag{21}$$

This assumption is reasonable since the maximum acceleration can be adjusted to match the reality.

Additionally, denote the feature point coordinates in I_r and I_v by $[u_1, v_1]^\top$ and $[u_2, v_2]^\top$, respectively. To ensure the target remains within the image range, the following condition must be satisfied:

$$|u_{1,k}| \leq \frac{w}{2}, \qquad |v_{1,k}| \leq \frac{h}{2}, \tag{22}$$

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where h and w represent the height and width of the image, respectively.

Definition 1 (Visibility): Due to the camera's limited FOV, (22) needs to be satisfied to make the target visible. This allows for the derivation of control inputs for the UAV based on the detected target's feature points.

Definition 2 (Long-term visibility): Long-term visibility means that the UAV maintains continuous visibility of the landing tag throughout the landing process. This is crucial in preventing situations where the tag may temporarily exceed the camera's FOV, thereby necessitating additional adjustments to make the tag back into the camera's FOV for a successful landing.

Since four corner points need to satisfy the constraints at the same time, then define $S = [u_{11}, v_{11}, \ldots, u_{14}, v_{14}]^\top$, and the state constraint set S_{set} can be given:

$$S_k \in S_{set} \subseteq \mathbb{R}^{8\times 1}. \tag{23}$$

Remark 1: The visibility of the target is critical for vision-based control, as it serves as a key input to the controller; otherwise control disruption may occur. In current research, efforts have been made to ensure target visibility through strategies, e.g. [25]; however, theoretical guarantees are lacking. Additionally, some studies directly address visibility constraints in the virtual image plane, e.g. [26], [27], but overlook the impact of UAV attitude changes. In reality, strict assurance of target visibility requires the definition of visibility constraints in the real image plane.

In order to analyze the visibility constraints shown in (22), the predicted feature points in the real image plane I_r need to be obtained. However, according to Section II-B, the system model is constructed based on $[u_{2l}, v_{2l}]^\top, l = \{1, 2, 3, 4\}$ in the virtual image plane I_v, which cannot be directly used to analyze the constraints. Therefore, it is necessary to map the feature points in I_v into those in I_r at first.

Based on the definition of image moments in (5), the corresponding feature point of $[u_c, v_c]^\top$ in the virtual image plane with an altitude of z^*, denoted by I_v^*, can be expressed as

$$u_c^* = f \cdot q_u, \quad v_c^* = f \cdot q_v. \tag{24}$$

Since the size of landing tag is known, then the length of it in I_v^* can be obtained, which is denoted by l_s^*. After that, the coordinates of four corner points in I_v^* are given as

$$\begin{cases} u_{2l}^* = u_c^* + l_d^* \cdot \cos\left(q_\psi + (l-1)\cdot \frac{\pi}{2}\right), \\ v_{2l}^* = v_c^* + l_d^* \cdot \sin\left(q_\psi + (l-1)\cdot \frac{\pi}{2}\right), \end{cases} \tag{25}$$

where $l_d^* = \frac{\sqrt{2}}{2} l_s^*$ and $l = \{1, 2, 3, 4\}$.

Furthermore, the corresponding feature points in the current virtual image plane with an altitude of z, are represented as

$$u_{2l} = \frac{u_{2l}^*}{q_h}, \quad v_{2l} = \frac{v_{2l}^*}{q_h}. \tag{26}$$

Denote the corresponding coordinates of $[u_{2l}, v_{2l}]^\top$ in $\mathcal{F}_c$ by $[x_l, y_l, z_l]^\top$, then

$$\begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = \mathrm{R}(\phi) \cdot \mathrm{R}(\theta) \cdot \begin{bmatrix} u_{2l} \\ v_{2l} \\ f \end{bmatrix}. \tag{27}$$

where $\mathrm{R}(\phi)$ and $\mathrm{R}(\theta)$ represent the rotation matrices of ϕ and θ, respectively. After that, $[u_1, v_1]^\top$ can be obtained as

$$\begin{bmatrix} u_{1l} \\ v_{1l} \end{bmatrix} = \frac{f}{z_2} \cdot \left(\mathrm{R}(\theta)^{-1} \cdot \mathrm{R}(\phi)^{-1} \cdot \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} \right)^{(1:2,\cdot)}, \tag{28}$$

where $(1:2, \cdot)$ represents the first two rows of a vector.

Remark 2: Due to the existence of unknown external disturbance E_j shown in (2), both $\theta_{k+i|k}$ and $\phi_{k+i|k}$ cannot be estimated in the prediction horizon, which results in the inability to obtain $[u_{1l}, v_{1l}]^\top$ based on (28).

Referring to [1], [32], the external disturbance is approximately proportional to the signed quadratic velocity. Therefore, when applying the velocity constraints (19) and acceleration constraints (20), both θ and ϕ will be limited in a small range according to (2). Besides, the amplitude of them, denoted by $\theta_{\max}, \phi_{\max} \in \mathbb{R}^+$, can be measured in advance.

In order to guarantee that the four corner points will be kept in the image range, the worst cases for them are taken into consideration. Concretely speaking, if $u_{2l} > 0$, the corner point is expected to exceed the image range faster when $\phi > 0$, and vice versa. Similarly, if $v_{2l} > 0$, the condition is $\theta < 0$.

To this end, we let

$$\begin{bmatrix} \bar{x}_l \\ \bar{y}_l \\ \bar{z}_l \end{bmatrix} = \mathrm{R}(\phi_m) \cdot \mathrm{R}(\theta_m) \cdot \begin{bmatrix} u_{2l} \\ v_{2l} \\ f \end{bmatrix}, \tag{29}$$

where

$$\begin{cases} \phi_m = \mathrm{sign}(u_{2l}) \cdot \phi_{\max}, \\ \theta_m = -\mathrm{sign}(v_{2l}) \cdot \theta_{\max}, \end{cases} \tag{30}$$

and

$$\mathrm{sign}(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} \tag{31}$$

After that, the corresponding coordinates of corner point in the real image plane, denoted by $[\bar{u}_{1l}, \bar{v}_{1l}]^\top, l = \{1, 2, 3, 4\}$, can be obtained based on (28). Therefore, the visibility constraints of (22) are reformulated as

$$\bar{u}_{1l,k+i|k} \leq \mathrm{sign}(u_{2l}) \cdot \frac{w}{2}, \quad \bar{v}_{1l,k+i|k} \leq \mathrm{sign}(v_{2l}) \cdot \frac{h}{2}, \tag{32}$$

which is abbreviated as

$$\bar{S}_k \in S_{set} \subseteq \mathbb{R}^{8\times 1}, \tag{33}$$

where $\bar{S} = [\bar{u}_{11}, \bar{v}_{11}, \ldots, \bar{u}_{14}, \bar{v}_{14}]^\top$.

In order to guarantee the convergence of the cost function (16), the following contractive constraint (CC) is given:

$$\mathrm{CC}_{k+N_p+1|k} \leq 0, \tag{34}$$

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where

$$\mathrm{CC}_{k+N_p+1|k} = J_r(\tilde{q}_{k+N_p+1|k}, U_{k+N_p|k}) - \lambda \cdot J_r(\tilde{q}_{k+1|k}, U_{k|k}), \tag{35}$$

and $\lambda \in (0, 1)$.

Note that the design of the contractive constraint (34) is to guarantee the stability of the closed-loop system, which will be proven in Theorem 1. Besides, the cost function (16) just focuses on the first N_p state. However, the constraint includes $\tilde{q}_{k+N_p+1|k}$ and $U_{k+N_p|k}$, which requires an additional forward prediction step based on the prediction horizon. Therefore, the control sequence of (18) is extended into

$$\mathbf{U}_k = \{U_{k|k}, U_{k+1|k}, \ldots, U_{k+N_p|k}\}. \tag{36}$$

To sum up, after combining (7), (16), (19), (20), (33), and (34), the optimization control problem can be formulated as follows:

$$\mathcal{P}: \quad \min_{\mathbf{U}_k} \quad J(\tilde{q}_{k|k}, \mathbf{U}_k) \tag{37a}$$

$$\text{s.t.} \quad q_{k+i+1|k} = \mathrm{f}_q(q_{k+i|k}, U_{k+i|k}), \tag{37b}$$

$$U_{k+i|k} \in U_{set}, \quad a_{k+i|k} \in a_{set}, \tag{37c}$$

$$\mathrm{CC}_{k+N_p+1|k} \le 0, \tag{37d}$$

$$\bar{S}_{k+i+1|k} \in S_{set}, \tag{37e}$$

where $i = 0, \ldots, N_p$.

B. Input-to-State Stability Analysis

Theorem 1: Based on the perturbed system of (8) with ξ belonging to the bounded set Ξ, the control input obtained from (37a)–(37e) guarantees the closed loop of the control system to be input-to-state stable for all $S \in S_{set}$.

Proof: According to (6) and (7), we have

$$\mathrm{f}_q(q_{k+1}, U) - \mathrm{f}_q(q_k, U) = \begin{bmatrix} 1 & -\omega_z^b \cdot dt & 0 & 0 \\ \omega_z^b \cdot dt & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot (q_{k+1} - q_k), \tag{38}$$

then

$$\|\mathrm{f}_q(q_{k+1}, U) - \mathrm{f}_q(q_k, U)\| \le \sqrt{1 + (\omega_z^b \cdot dt)^2} \cdot \|q_{k+1} - q_k\|. \tag{39}$$

Therefore, subject to the constraints (19) and (23), the nominal system (7) is locally Lipschitz in q for all $S_k \in S_{set}$, and the Lipschitz constant is

$$C_f = \sqrt{1 + (\bar{\omega}_z^b \cdot dt)^2}, \tag{40}$$

where $\bar{\omega}_z^b$ represents the maximum of ω_z^b.

In terms of the stage cost $J_r(\tilde{q}, U)$, there exists

$$J_r(\tilde{q}_{k+1}, U) - J_r(\tilde{q}_k, U) \le 2\gamma \cdot \sigma_{\max}(Q) \cdot \|\tilde{q}_{k+1} - \tilde{q}_k\|, \tag{41}$$

where $\gamma = \max(\|\tilde{q}\|)$ and $\sigma_{\max}(Q)$ represents the maximum singular value of Q. Therefore, J_r is locally Lipschitz in S_{set} with Lipschitz constant $C_{J_r} = 2\gamma \cdot \sigma_{\max}(Q)$.

It is obvious from (8) that

$$\|\hat{q}_{k|k} - \hat{q}_{k|k-1}\| \le \bar{\xi}. \tag{42}$$

Substituting (14) into (42) can yield

$$\|\tilde{q}_{k|k} - \tilde{q}_{k|k-1}\| \le \bar{\xi}. \tag{43}$$

Through iterative derivation, the following formulation can be obtained:

$$\|\tilde{q}_{k+N_p-1|k} - \tilde{q}_{k+N_p-1|k-1}\| \le (C_f)^{N_p-1} \cdot \bar{\xi}. \tag{44}$$

Comprehensively considering (41) and (44), then

$$J_r(\tilde{q}_{k+i|k}, U) - J_r(\tilde{q}_{k+i|k-1}, U) \le C_{J_r} \cdot \|\tilde{q}_{k+i|k} - \tilde{q}_{k+i|k-1}\| \le C_{J_r} \cdot (C_f)^i \cdot \bar{\xi}. \tag{45}$$

Denote the optimal control sequence at time-step $k-1$ by $\mathbf{U}_{k-1}^*$, then the first N_p elements of feasible control sequence $\mathbf{U}_k$ at time-step k can be obtained as follows:

$$U_{k+i|k} = U_{k+i|k-1}^*, \quad i = 0, \ldots, N_p - 1 \tag{46}$$

Denote the optimal cost in time-step $k-1$ by J_{k-1}^* and the cost arising from $\mathbf{U}_k$ at time-step k by J_k, then the error between J_{k-1}^* and J_k is

$$\begin{aligned} \Delta J &= J_k - J_{k-1}^* \\ &= \sum_{i=1}^{Np-1} \Big(J_r(\tilde{q}_{k+i|k}, U_{k+i-1|k}) - J_r(\tilde{q}_{k+i|k-1}, U_{k+i-1|k-1}^*) \Big) \\ &\quad - J_r(\tilde{q}_{k|k-1}, U_{k-1|k-1}^*) + J_r(\tilde{q}_{k+N_p|k}, U_{k+N_p-1|k}). \end{aligned} \tag{47}$$

Substituting (45) into (47), we have

$$\begin{aligned} \Delta J \le & \sum_{i=1}^{Np-1} C_{J_r} \cdot (C_f)^i \cdot \bar{\xi} \\ & + \Big(J_r(\tilde{q}_{k+N_p|k-1}, U_{k+N_p-1|k-1}^*) - J_r(\tilde{q}_{k|k-1}, U_{k-1|k-1}^*) \Big) \\ & + \Big(J_r(\tilde{q}_{k+N_p|k}, U_{k+N_p-1|k}) - J_r(\tilde{q}_{k+N_p|k-1}, U_{k+N_p-1|k-1}^*) \Big) \\ \le & \sum_{i=1}^{Np} C_{J_r} \cdot (C_f)^i \cdot \bar{\xi} \\ & + \Big(J_r(\tilde{q}_{k+N_p|k-1}, U_{k+N_p-1|k-1}^*) - J_r(\tilde{q}_{k|k-1}, U_{k-1|k-1}^*) \Big) \end{aligned} \tag{48}$$

Furthermore, according to (34), there exists

$$\Delta J \le \sum_{i=1}^{Np} C_{J_r} \cdot (C_f)^i \cdot \bar{\xi} - (1 - \lambda) \cdot J_r(\tilde{q}_{k|k-1}, U_{k-1|k-1}). \tag{49}$$

Finally, since the optimal solution J_k^* is not greater than J_k, it yields

$$J_k^* - J_{k-1}^* \le \Delta J. \tag{50}$$

Therefore, according to Theorem 2 in [33], with the aid of control input obtained from (37a), the closed loop of the control system is input-to-state stable. □

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Furthermore, in order to guarantee that the state $\tilde{q}$ and the corresponding control input U will reach and then stay within a small region around 0, which is defined as $\mathcal{Q} = \{\tilde{q} \in \mathbb{R}^{4\times 1}, U \in U_{set} \mid J_r(\tilde{q}, U) \leq \varepsilon\}, \varepsilon \in \mathbb{R}^+$, we need to find the bound $\bar{\xi}$. For simplicity of notation, $J_r(\tilde{q}, U)$ is abbreviated as $J_r(\tilde{q})$ in the subsequent discussions.

Theorem 2: Consider the perturbed system of (8), the control input provided by (37a)–(37e) guarantees that the state $\tilde{q}$ and control input U will be driven to $\mathcal{Q}$ and then stabilized within it, iff

$$\bar{\xi} \leq \frac{(1-\lambda)\cdot\varepsilon}{C_{J_r}\cdot(C_f)^{N_p}}, \tag{51}$$

where $C_f = \sqrt{1+(\bar{\omega}_z^b \cdot dt)^2}$, $C_{J_r} = 2\gamma \cdot \sigma_{\max}(Q)$.

Proof: According to (35) and (45), we have

$$J_r(\tilde{q}_{k+N_p|k}) \leq C_{J_r}\cdot(C_f)^{N_p}\cdot\bar{\xi} + \lambda \cdot J_r(\tilde{q}_{k|k-1}). \tag{52}$$

Actually, $\tilde{q}_{k+N_p|k}$ can be regarded as the perturbed state of $\tilde{q}_{k+N_p|k-1}$. Therefore, when $J_r(\tilde{q}_{k|k-1}) \in \mathcal{Q}$ is satisfied, it is expected that $J_r(\tilde{q}_{k+N_p|k}) \in \mathcal{Q}$, and we get

$$C_{J_r}\cdot(C_f)^{N_p}\cdot\bar{\xi} + \lambda\cdot\varepsilon \leq \varepsilon, \tag{53}$$

which yields (51).

Furthermore, based on (52), when $J_r(\tilde{q}_{k|k-1}) \notin \mathcal{Q}$, i.e. $J_r(\tilde{q}_{k|k-1}) > \varepsilon$, we have

$$J_r(\tilde{q}_{k+N_p|k}) \leq (1-\lambda)\cdot\varepsilon + \lambda\cdot J_r(\tilde{q}_{k|k-1}) < J_r(\tilde{q}_{k|k-1}) \tag{54}$$

which leads to

$$J_r(\tilde{q}_{k+i\cdot N_p}) < J_r(\tilde{q}_{k+N_p}) < J_r(\tilde{q}_{k|k-1}), \quad i \in \mathbb{Z}^+ \tag{55}$$

This indicates that the state $\tilde{q}$ and U will converge to $\mathcal{Q}$ and then be stabilized within it. Then the proof is completed. □

Remark 3: The perturbed term ξ_k in (8) consists of both model uncertainties and external disturbances, such as winds. Due to the influence of ξ_k, the state q_k is unable to converge to q_d, which results in a steady-state error. Moreover, when ξ_k is bounded as described in (51), the landing error of the UAV will be constrained within a domain belonging to $\mathcal{Q}$.

C. Two-Stage Optimization Control

When a rotor UAV initiates descent from a high altitude using vision sensors, it requires a large tag for initial detection to guide the UAV to a low altitude. However, if the UAV descends to a low altitude, the large tag may surpass the FOV of the camera as the UAV continues its descent, resulting in the inability to provide accurate landing guidance. To address this issue, the use of nested small tag can offer terminal guidance, facilitating the accurate landing of the UAV.

Remark 4: Based on two different types of AprilTags, the optimal problem can be divided into two parts. Define the process when only the big tag is detected as the first stage, while the process when the small tag is detected as the second stage. When the rotor UAV begins to land, it is expected to move quickly at the first stage for a fast landing, and to adjust slowly at the second stage for an accurate landing.

Denote the different control constraint sets at the two stages by U_1 and U_2, respectively. Based on (19), there exist

$$U_1 \in U_{1,set}, \quad U_2 \in U_{2,set}, \tag{56}$$

which can be rewritten as

$$U_{1,\min} \leq U_1 \leq U_{1,\max}, \quad U_{2,\min} \leq U_2 \leq U_{2,\max}, \tag{57}$$

where

$$U_{1,\min} = [-\mu_x, -\mu_y, -\mu_z, -\mu_\omega]^\top, \tag{58}$$

$$U_{1,\max} = [\mu_x, \mu_y, 0, \mu_\omega]^\top, \tag{59}$$

$$U_{2,\min} = [-\nu_x, -\nu_y, -\nu_z, -\nu_\omega]^\top, \tag{60}$$

$$U_{2,\max} = [\nu_x, \nu_y, 0, \nu_\omega]^\top, \tag{61}$$

and $\mu_i, \nu_i \in \mathbb{R}^+, i \in \{x, y, z, \omega\}$. Besides, according to Remark 4, we let $\mu_i > \nu_i$ to make the UAV move quickly at the first stage and slowly at the second stage.

In practical, compared with the optimality of the optimization control problem (37a), the feasibility of its solution needs to be prioritized. Considering that the nonlinear constraints (37d) and (37e) may lead to the infeasible solution, a soft constraint method based on *max* function is introduced to solve the problem. Combining (16), (22), and (34), the cost function with soft constraints can be rewritten as

$$\begin{aligned} J_{soft}(\tilde{q}_{k|k}, \mathbf{U}_k) = J(\tilde{q}_{k|k}, \mathbf{U}_k) + \beta_1 \cdot \max\left\{CC_{k+N_p+1|k}, 0\right\} \\ + \sum_{i=1}^{Np+1}\sum_{l=1}^{4}\Big(\beta_2 \cdot \max\left\{\bar{u}_{1\,l,k+i|k}^2 - w_u, 0\right\} \\ + \beta_3 \cdot \max\left\{\bar{v}_{1\,l,k+i|k}^2 - w_v, 0\right\}\Big). \end{aligned} \tag{62}$$

Note that $\beta_1, \beta_2, \beta_3 \in \mathbb{R}^+$. Besides, we let $w_u < \frac{w^2}{4}$ and $w_v < \frac{h^2}{4}$, which is to ensure that even though the constraints

$$u_{1,k+i|k}{}^2 - w_u \leq 0, \quad v_{1,k+i|k}{}^2 - w_v \leq 0$$

are slightly violated, the landing tag will still within the image range described by (22).

Denote the desired altitude at the first stage and second stage by z_1^* and z_2^* ($z_1^* > z_2^* > 0$), respectively. Correspondingly, the image moment vector (5) is written as q_1 and q_2 at the two stages, respectively. After that, the optimization control problem at the first stage can be formulated as

$$\mathcal{P}_1: \quad \min_{\mathbf{U}_{1,k}} \quad J_{soft}(\tilde{q}_{1,k|k}, \mathbf{U}_{1,k}) \tag{63a}$$

$$\text{s.t.} \quad q_{1,k+i+1|k} = \mathrm{f}_q(q_{1,k+i|k}, U_{1,k+i|k}), \tag{63b}$$

$$U_{1,k+i|k} \in U_{1,set}, \quad a_{k+i|k} \in a_{set}, \tag{63c}$$

and the optimization control problem at the second stage can be formulated as

$$\mathcal{P}_2: \quad \min_{\mathbf{U}_{2,k}} \quad J_{soft}(\tilde{q}_{2,k|k}, \mathbf{U}_{2,k}) \tag{64a}$$

$$\text{s.t.} \quad q_{2,k+i+1|k} = \mathrm{f}_q(q_{2,k+i|k}, U_{2,k+i|k}), \tag{64b}$$

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Algorithm 1: Robust IBVS-MPC Method.

Input: (u_{2l}, v_{2l}), $l = \{1, 2, 3, 4\}$
Output: the control input U_k^*

```
Initialize control sequence U_0 = 0, set k = 0.
Measure θ_max and φ_max to formulate (32).
while q̃_{2,k} ∉ Q_2 do
    Update q_{2,k} based on (7).
    if the small tag is detected then
        if U_{1,k} ∈ U_{2,set} then
            Solve P_2 to obtain U_{2,k}, set U_k^* = U_{2,k|k}.
        else
            Solve P_1 to obtain U_{1,k}, set U_k^* = U_{1,k|k}.
        end
    else
        if the big tag is detected then
            Solve P_1 to obtain U_{1,k}, set U_k^* = U_{1,k|k}.
        else
            Set U_k^* according to Remark 5.
        end
    end
    k = k + 1.
end
Shut down the motors.
```

$$U_{2,k+i|k} \in U_{2,set}, \quad a_{k+i|k} \in a_{set}, \tag{64c}$$

where $i = 0, \ldots, N_p$.

Furthermore, q_d at the first stage is rewritten as

$$q_{d,1} = [0, 0, 1 - \epsilon, 0]^\top, \tag{65}$$

where $\epsilon \in (0, 1)$ and is closer to 0. This is to make $V_z^b < 0$ when the UAV reaches the given desired altitude of $z_1^* > 0$, so that it can continue to descend for the smooth transition from the first stage to the second stage.

According to Theorem 1, the control systems at the two stages are input-to-state stable. That is, with the aid of $\mathcal{P}_1$, the UAV will descend to an altitude of z_1^* and then the small tag can be detected. After that, since the control process is discrete and $V_z^b \leq 0$, then the small tag can be detected from then on. Besides, in order to guarantee that the feasible control sequence of $\mathcal{P}_2$ can be obtained from the optimal solution of $\mathcal{P}_1$ like (46), the condition $U_1 \in U_{2,set}$ needs to be satisfied to switch the optimization control problem.

Actually, both the big tag and small tag will not be detected at some time during the landing of the UAV. On the one hand, this may be caused by the inability of the detection algorithm to recognize them. On the other hand, the tags may have exceeded the FOV. To solve the problem, the strategy is given as follows:

Remark 5: If both the tags are not detected since time-step $k + 1$, which will last to time-step $k + N, N \in \mathbb{Z}^+$, then we set $U_{k+i}^* = U_k^*, i = 1, \ldots, N$. This is to avoid the case that tags are within the FOV while unable to be detected, which is beneficial to ensure the smoothness of landing. Once the tags still cannot be detected since time-step $k + N + 1$, then they are probably out of sight, and we set $U_{k+N+1}^* = 0$ to keep safe.

Finally, the control input is chosen as $U_k^* = U_{1,k|k}$ (or $U_{2,k|k}$). Since the landing platform is moving, it is unable for the UAV to land accurately at the center of the landing tag. Therefore, we define a small region around the origin: $\mathcal{Q}_2 = \{\tilde{q} \in \mathbb{R}^{4\times 1} | \|\tilde{q}\| \leq \varepsilon_2\}$, $\varepsilon \in \mathbb{R}^+$. Once $\tilde{q}_{2,k} \in \mathcal{Q}_2$ is satisfied, the motors are shut down to make the UAV fall vertically. The proposed robust MPC-IBVS method is outlined in Algorithm 1.

IV. Numerical Simulations and Results

A. Comparative Experiments

In this set of simulations, the parameters are set as follows: $g = 9.8\ \mathrm{m/s^2}$, $w = 1280$, $h = 720$, $Q = diag([10, 10, 10, 10])$, $R = diag([1, 1, 1, 1])$ and $\lambda = 0.8$. In terms of the square landing tag, the initial coordinates of four corner points in the inertial frame are $[1, -1, 0]^\top$, $[1, 1, 0]^\top$, $[-1, 1, 0]^\top$ and $[-1, -1, 0]^\top$. Simultaneously, the initial position of the UAV in the inertial frame are $[6.05, 1.5, 8]^\top$, while its attitude is $\psi = \frac{\pi}{2}$, $\phi = 0$ and $\theta = 0$. According to the perspective projection, the corresponding coordinates of four corner points in $\mathcal{F}_i$ are $[-339.0, 167.82]^\top$, $[-339.0, 33.56]^\top$, $[-473.25, 33.56]^\top$ and $[-473.25, 167.82]^\top$, respectively. The desired coordinates of four corner point in $\mathcal{F}_i$ are set as $[235, -235]^\top$, $[-235, -235]^\top$, $[-235, 235]^\top$ and $[235, 235]^\top$ in turn. Note that $U_{\min} = [-1, -1, -0.8, -0.2]^\top$, $U_{\max} = [1, 1, 0, 0.2]^\top$, $a_{\min} = [-0.4, -0.4, -0.4, -0.4]^\top$ and $a_{\max} = [0.4, 0.4, 0.4, 0.4]^\top$, where the units of linear velocity and angular velocity are (m/s) and (rad/s), respectively. In addition, we assume that the initial state of the UAV is static, the external disturbance is proportional to the signed quadratic velocity, i.e. $\frac{k_d}{m}$ in [1] is set to 0.5, and $dt = 0.1$ s, $N_p = 10$. The optimization problem is solved by using the *fmincon* function in MATLAB.

The first simulation directly considers the visibility constraints in the virtual image plane as in [27], i.e. $|u_2| < \frac{w}{2}$ and $|v_2| < \frac{h}{2}$. Four corner points of the landing tag are denoted by p_1, p_2, p_3, p_4, and the black dashed lines represent the boundaries of the image. Fig. 3(a) shows the time evolution of four corner points in the virtual image plane. It is evident that these four points consistently remain within the virtual image range. However, p_4 has moved beyond the bounds of the real image plane at about 2 s, as depicted in Fig. 3(b). Additionally, the alterations in UAV attitude, as shown in Fig. 3(c), account for the disparities between Fig. 3(a) and (b). To mitigate control disruptions stemming from brief target loss, target state prediction is typically employed. However, owing to model inaccuracies, the prediction errors accumulate gradually over time, necessitating a small prediction horizon. If the target fails to re-enter the image range within the specified horizon, it is deemed to have deviated from the image range, prompting the UAV to maintain static for safety reasons. Here we adopt 10 prediction steps, and the corresponding results are shown in the partial enlarged view of the area inside the orange ellipse in Fig. 3(b). Evidently, p_4 is still out of sight during the prediction horizon, and the trend of v_1 still appears to be increasing. Subsequently, the UAV velocity

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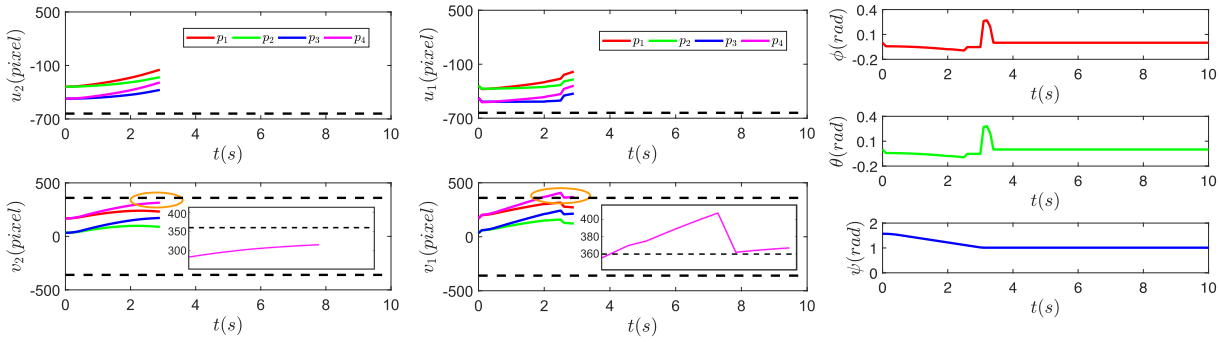


Fig. 3. Simulation results for considering the visibility constraint directly in the virtual image plane, e.g. [27].

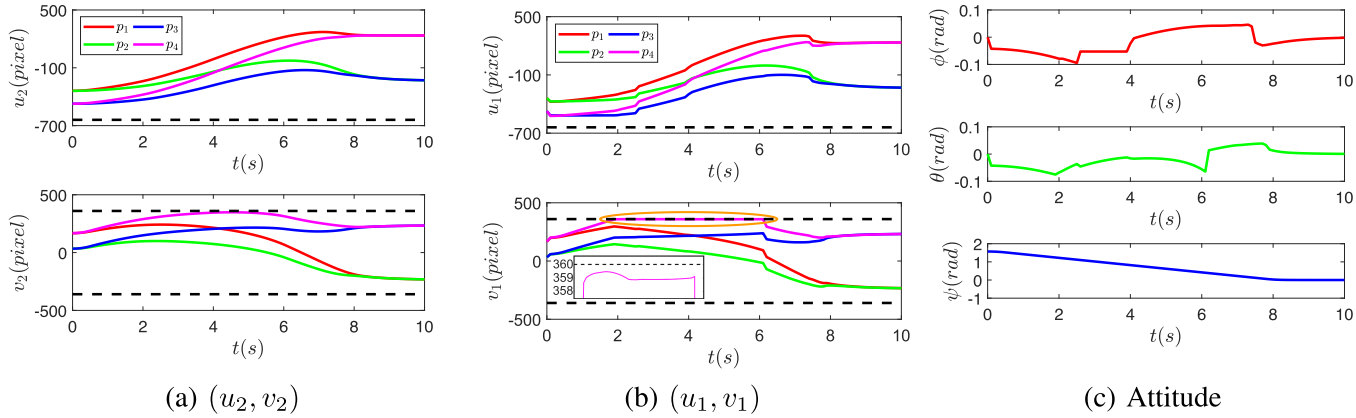


Fig. 4. Simulation results for the method proposed in this work.

is set to zero, which results in attitude oscillations at about 3 s, as indicated in Fig. 3(c).

In comparison, the second simulation validates the method proposed in this work, with the corresponding results presented in Fig. 4. It can seen from Fig. 4(b) that the four corner points consistently remain within the image range. Based on the partial enlarged view, although v_1 of p_4 approaches the boundary closely, the point never exceeds the image range. By comparing Fig. 3(c) with Fig. 4(c), it can be observed that $|\theta|$ in the latter reaches about 0.075 rd at 1.9 s and subsequently decreases, thereby ensuring that v_1 does not surpass $\frac{h}{2}$. Similarly, $|\theta|$ in the former also reaches about 0.075 rd at 1.9 s. However, it continues to increase from 1.9 s to 2.5 s, leading to the target exceeding the image range. Therefore, our method ensures long-term target visibility for the continuity of control. Besides, as the attitude converges to zero at 10 s, the virtual corner points coincide with the corresponding real corner points, with both converging to the desired positions, as demonstrated in Fig. 4(a) and (b).

Furthermore, to better elucidate the disparity between the two simulations, the UAV velocity is provided in Fig. 5. The solid lines represent the UAV velocity in the first simulation, while the dashed lines depict the velocity in the second simulation. It is noteworthy that θ is directly influenced by the acceleration

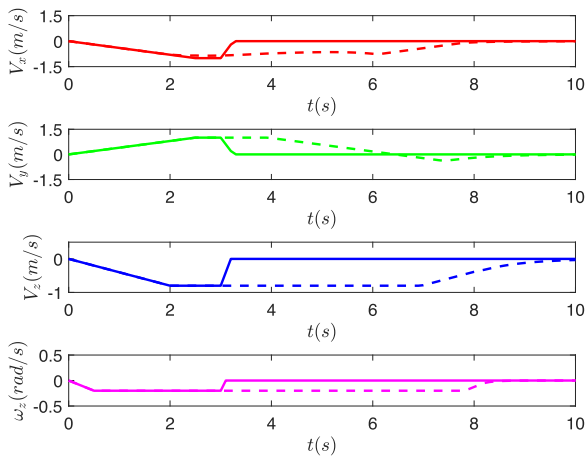
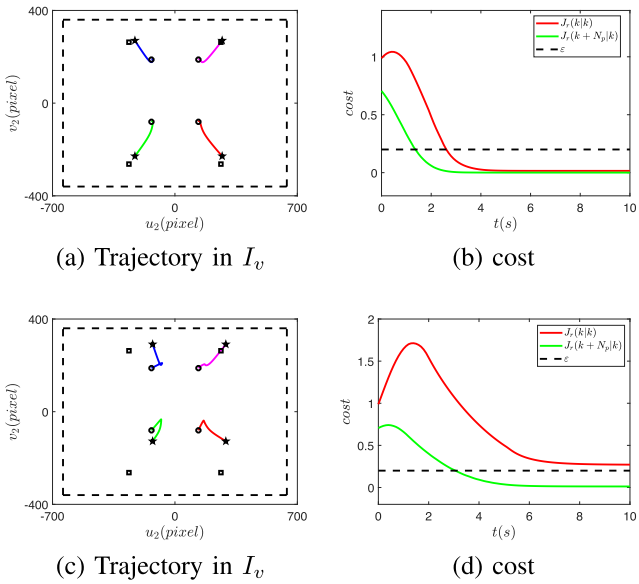


Fig. 5. UAV Velocity in the two simulations.

in the x_b-axis, i.e. the change of V_x in Fig. 5. We can see that the magnitude of V_x represented by the solid line continues to increase during [2, 2.5] s, consequently contributing to the rise in $|\theta|$ during the period in Fig. 3(c). In contrast, the magnitude of V_x depicted by the dashed line tends to remain relatively constant and subsequently decreases after 2 seconds. This behavior leads

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(a) Trajectory in I_v (b) cost

(c) Trajectory in I_v (d) cost

Fig. 6. (a) and (b) simulation results when $\xi_a \approx 0.017$. (c) and (d) simulation results when $\xi_b \approx 0.0565$.

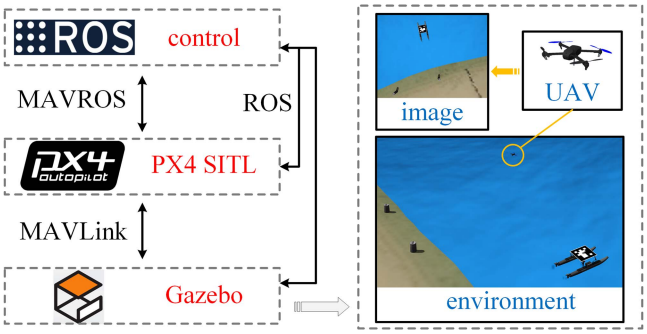


Fig. 7. SITL simulation setup.

to a reduction in $|\theta|$ in Fig. 4(c), thereby ensuring the retention of the target within the image range.

B. Robustness to Disturbance

In this set of simulations, by adjusting the initial pose of the UAV and keeping the position of four corner points unchanged, the corresponding four coordinates in $\mathcal{F}_i$ are changed into: $[134.26, -80.55]^\top$, $[-134.26, -80.55]^\top$, $[-134.26, 187.96]^\top$ and $[134.26, 187.96]^\top$. Simultaneously, choose $\gamma = 1$, $\varepsilon = 0.2$, $\lambda = 0.8$. Based on (40) and (41), there exist $C_f \approx 1.0002$, $C_{J_r} = 2$. After that, $\bar{\xi} \approx 0.02$ can be obtained according to (51). Besides, other parameters remain unchanged compared to those in the above set of simulations.

In order to verify the robustness of the closed-loop system, we add a constant disturbance on the system model (8). More specifically, let $\Delta x, \Delta y, \Delta z = 0.015$ in the first case, and we have $\xi_a \approx 0.017$ according to (9)–(13). Similarly, we let $\Delta x, \Delta_y, \Delta z = 0.05$ in the second case, which leads to $\xi_b \approx 0.0565$. The results are shown in Fig. 6. By comparing the trajectories of four corner points in Fig. 6(a) and (c), it is obvious that the final positions (expressed by pentagrams) of the four corner points in the first simulation are close to the desired positions (expressed by squares) than those of the second simulation. In order to quantitatively analyze the differences, the time evolution of the cost J_r are provided in Fig. 6(b) and (d). Note that $J_r(k+i|k)$ is the abbreviation of $J_r(\tilde{q}_{k+i|k}, U_{k+i-1|k})$. It can be observed that $J_r(k+N_p|k)$ derived from the nominal model will finally be less than ε in both simulations. Besides, since $\xi_a < \bar{\xi}$, the cost $J_r(k|k)$ reflecting the real state will finally be less than ε as well shown in Fig. 6(b). However, due to $\xi_b > \bar{\xi}$ in the second case, then $J_r(k|k)$ in Fig. 6(d) will be not less than ε. This indicates that when $\xi > \bar{\xi}$, the state $\tilde{q}$ and control input U is unable to be stabilized within $\mathcal{Q}$.

V. SITL Simulations and Results

A. Simulation Setup

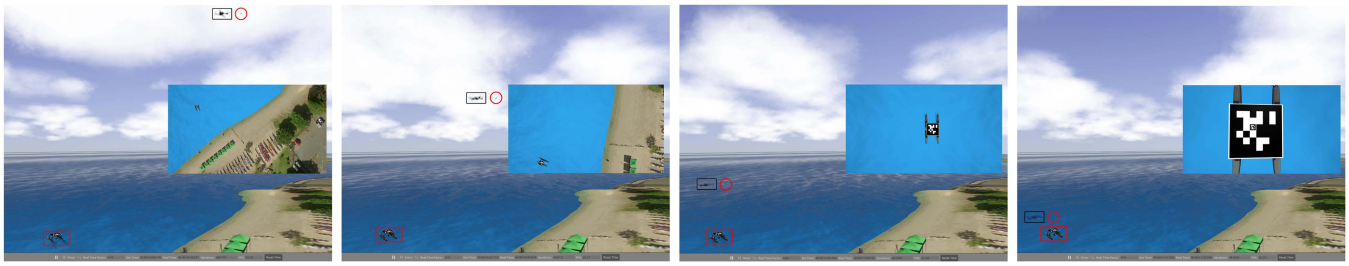
A high-fidelity simulation system is developed to evaluate the performance of our algorithm. Actually, the system is built based on XTdrone platform [34], which is open-source and contributed by our team. Besides, the *CppAD* toolbox is used to solve the optimization problem with C++ code. Notably, the algorithm verified by the system can be directly transferred to the field experiments, thereby improving the efficiency.

The simulation system is illustrated in Fig. 7, mainly composed of three parts: Robot Operation System[2] (ROS), PX4[3] and Gazebo.[4] ROS serves as a publish-subscribe communication framework and plays a key role in processing the control program and facilitating communication with PX4 and Gazebo. PX4, an autopilot software platform, operates independently of any hardware. By utilizing MAVROS[5], ROS messages can be converted into MAVLink[6] format, enabling PX4 to process the messages and calculate the propeller speeds required for controlling the rotor UAV. Simultaneously, the UAV can obtain its own state through MAVROS as well.

Gazebo is a simulator that facilitates the creation of an experimental environment, encompassing various models such as a quadrotor UAV, a boat with a pasted tag, winds and waves. The rotor UAV is equipped with a downward camera, featuring a FOV of 100° and a resolution of 1280 × 720 pixels. The wind and wave models employed in this work refer to [35]. Specifically, we set the constant mean wind speed to 0, and the standard deviation of zero-mean variable wind speed is set to 2. Additionally, the time constant is set to 1 and the simulation time step is set to 10 Hz. Regarding the wave model, we define its magnitude as 0.5 m, and the period is set to 5 s.

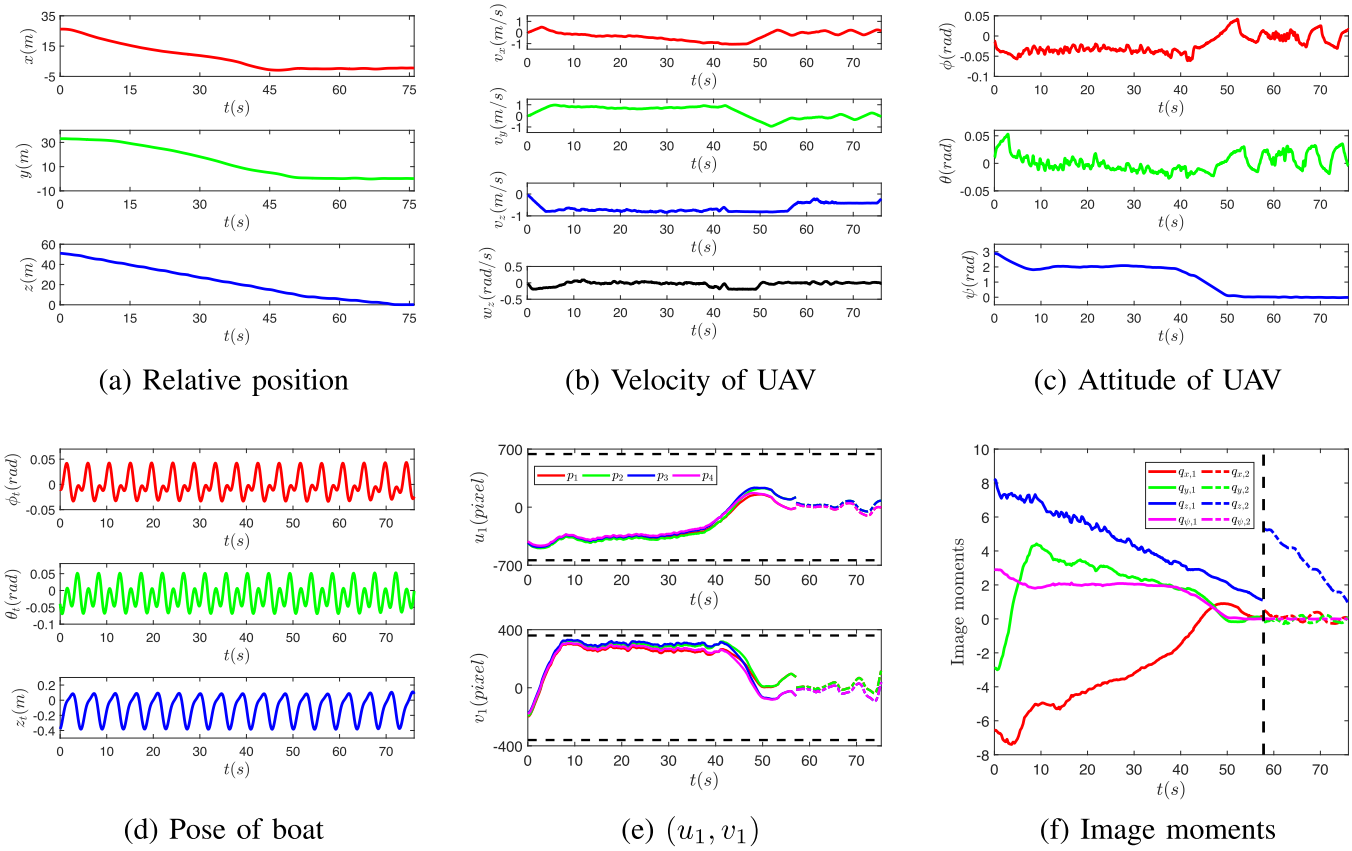
[2]https://www.ros.org/
[3]https://px4.io/
[4]https://gazebosim.org/home
[5]http://wiki.ros.org/mavros
[6]https://mavlink.io/

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(a) t = 0 s (b) t = 30 s (c) t = 60 s (d) t = 76 s

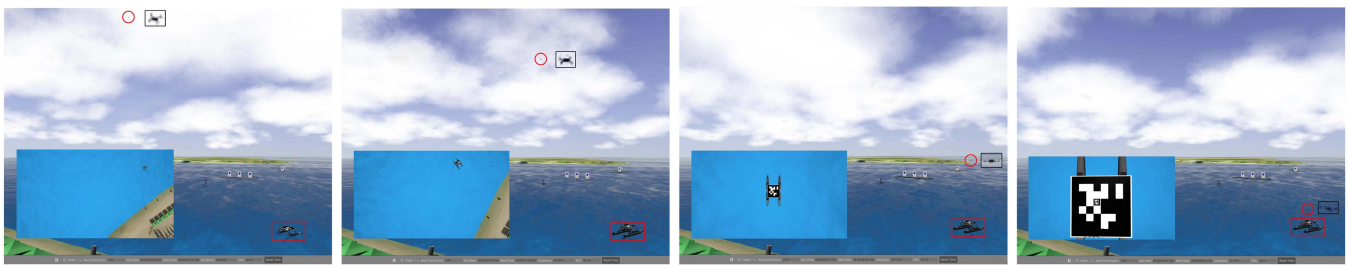
Fig. 8. Snapshots of the landing in the first case of SITL simulations. The red circle represents the rotor UAV, while the red rectangle represents the boat. Note that the black rectangle next to the red circle is the partial enlarged view of the UAV. Besides, the picture inside is the captured image of the downward camera.



(a) Relative position (b) Velocity of UAV (c) Attitude of UAV

(d) Pose of boat (e) (u_1, v_1) (f) Image moments

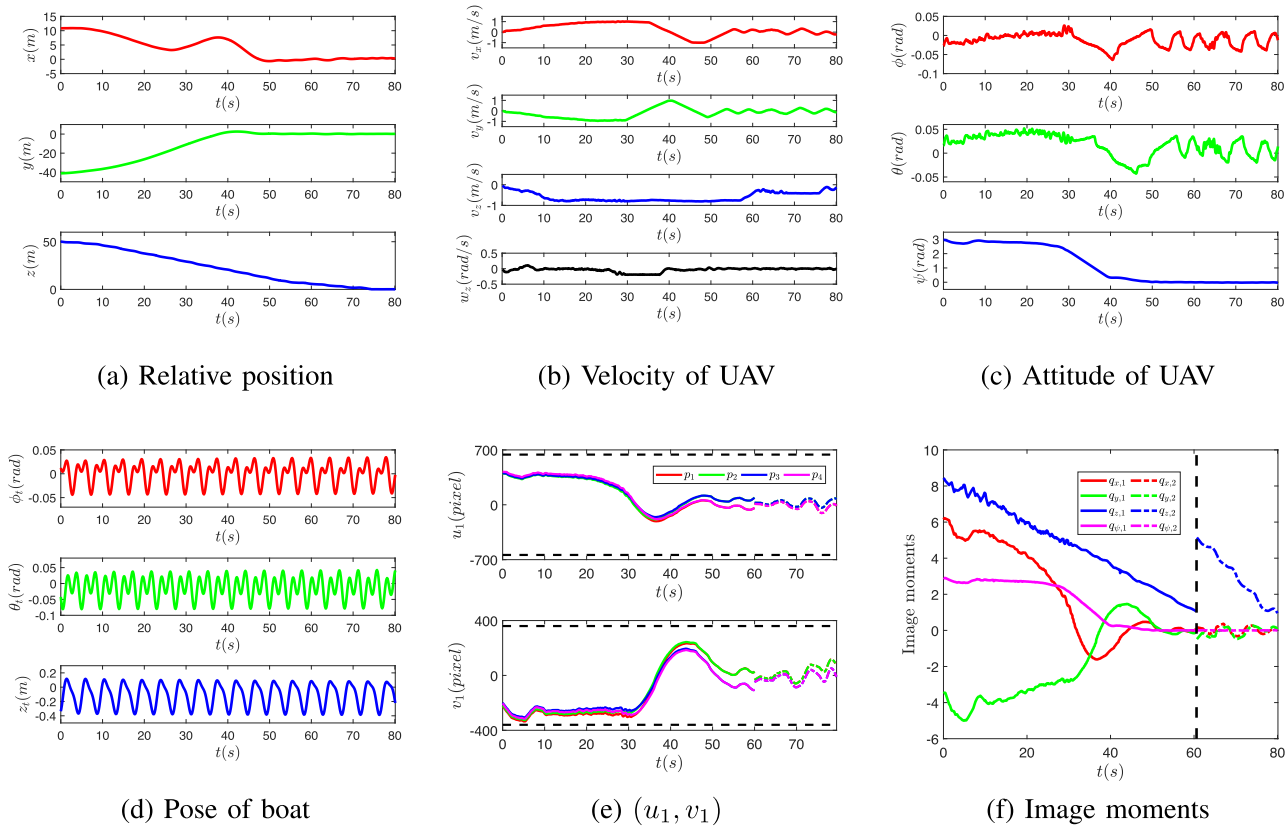
Fig. 9. Landing results corresponding to the first case in SITL simulations.



(a) t = 0 s (b) t = 30 s (c) t = 60 s (d) t = 80 s

Fig. 10. Snapshots of the landing in the second case of SITL simulations.

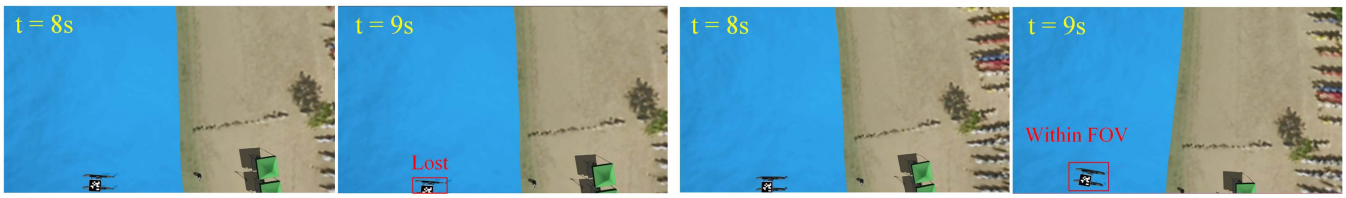
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(a) Relative position (b) Velocity of UAV (c) Attitude of UAV

(d) Pose of boat (e) (u_1, v_1) (f) Image moments

Fig. 11. Landing results corresponding to the second case in SITL simulations.



(a) Consider visibility constraint in I_v, e.g. [27] (b) Consider visibility constraint with our method

Fig. 12. Images captured by the downward camera in different time instants.

B. Experimental Tests

The control constraint sets (58)–(61) in the simulation are $U_{1,\min} = [-1, -1, -0.8, -0.2]^\top$, $U_{1,\max} = [1, 1, 0, 0.2]^\top$, $U_{2,\min} = [-0.4, -0.4, -0.4, -0.2]^\top$, $U_{2,\max} = [0.4, 0.4, 0, 0.2]^\top$. The other parameters are set as follows: $\beta_1 = \beta_2 = \beta_3 = 500$, $\lambda = 0.8$, $\epsilon = 0.05$, $\phi_{\max} = 0.1$, $\theta_{\max} = 0.1$, $w_u = 620$, $w_v = 340$. Besides, l_s^* of (25) is 85 pixel in the first stage, while is 40 pixel in the second stage. In order to verify the feasibility of our method, two cases with different initial relative position of the landing tag with respect to the UAV are considered.

In the first case, the initial altitude of the UAV relative to the landing tag is about 50 m, and the initial position of the tag lies in the upper left corner of the camera image, shown as Fig. 8(a). Besides, Fig. 8 displays the snapshots of different instants during the landing, and the embedded picture in each subfigure represents the captured image by the downward camera. It can be seen that the tag will be kept near the image center at about 76 s, thereby achieving an accurate landing.

To analyze the landing results quantitatively, more details are given in Fig. 9. The time evolution of the relative position is shown in Fig. 9(a). Both x and y will converge to 0 at about 50 s. Simultaneously, the change rate of z before 57 s is fast, which corresponds to the first stage. However, it is slow after 57 s to make minor adjustment concurrently keeping the tag inside the FOV. Furthermore, the time evolution of velocity for the UAV is given in Fig. 9(b), which accounts for the change of relative position in Fig. 9(a). Note that v_z before 55 s is nearly kept at about -0.8 m/s for a fast landing, and after smooth transition, it is changed into about -0.4 m/s for accurate landing. Fig. 9(c) shows the change of attitude for the rotor UAV. It is obvious

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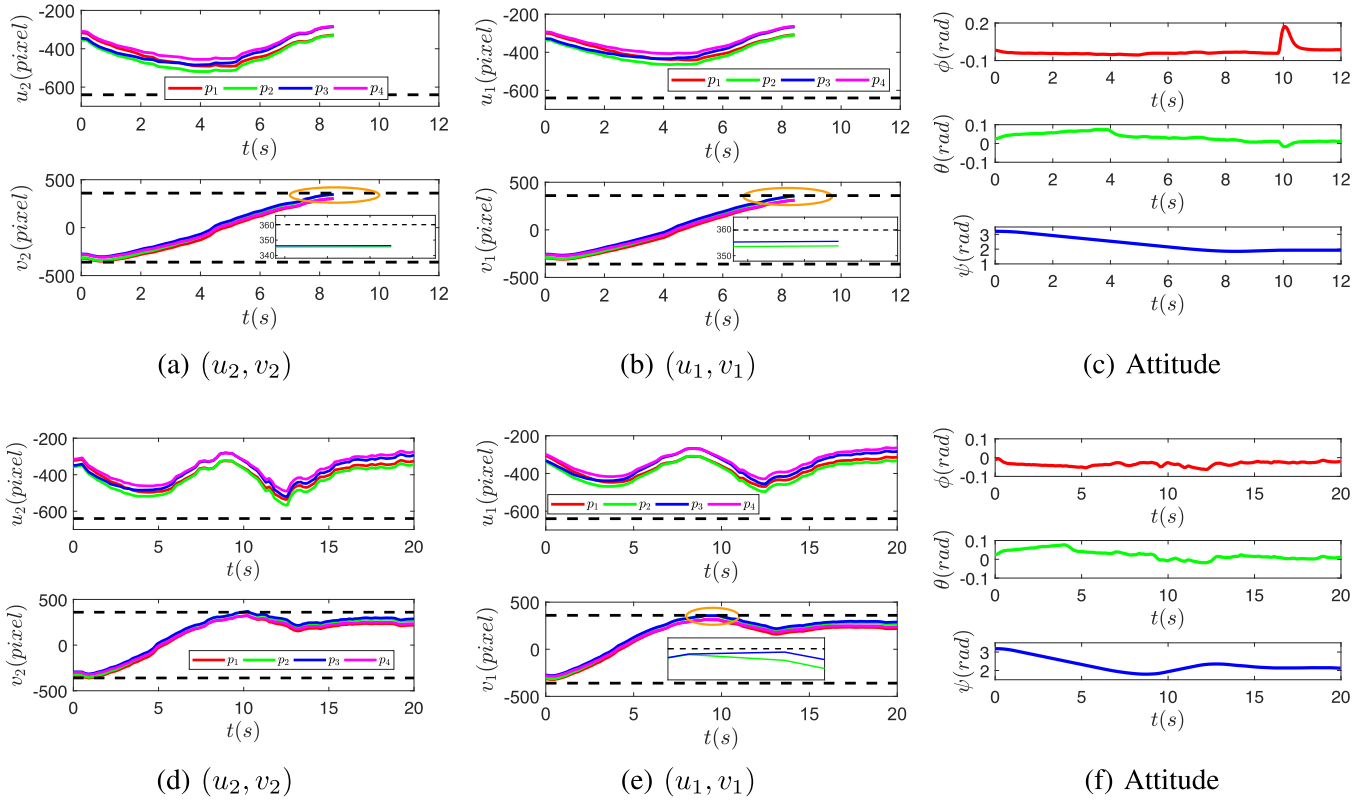


Fig. 13. (a), (b), and (c) landing results when considering the visibility constraints directly in I_v, e.g. [27]. (d), (e), and (f) landing results when considering the visibility constraint with our method.

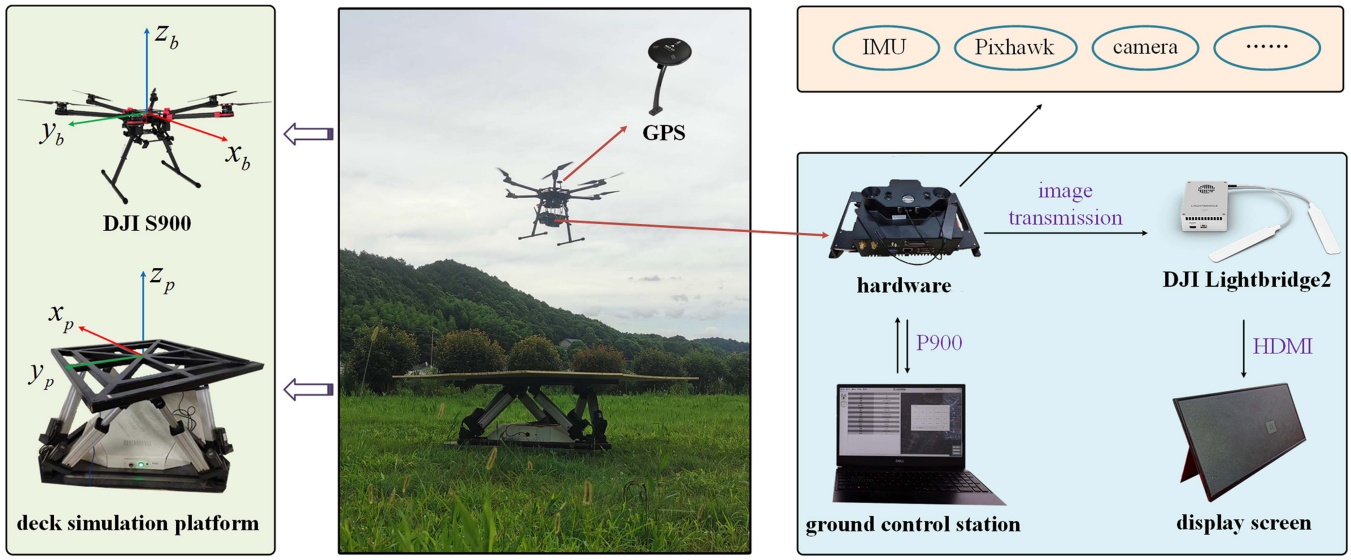


Fig. 14. Field experimental setup.

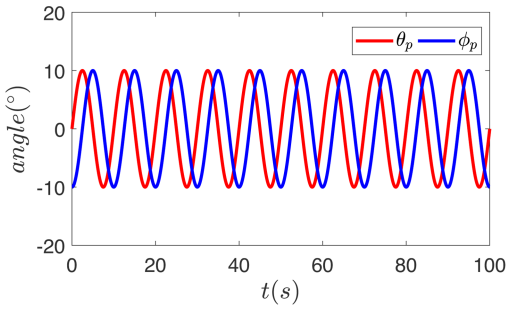
that both θ and ϕ are less than $\theta_{\max}$ and $\phi_{\max}$, respectively. Moreover, ψ almost keeps unchanged between [10, 40] s, which is to prevent the landing tag from exceeding the FOV, e.g. the snapshot corresponding to $t = 30$ s in Fig. 8. It can be seen that the tag has reached the bottom side of the camera image, and it will soon exceed the image range once ψ continues to descend.

In order to reflect the motion of the tag, its pose is shown in Fig. 9(d). Due to the effect of waves, the volatility in the z direction of Gazebo world is about 0.5 m, which accounts for the fluctuation of the error q_z in Fig. 9(f). Simultaneously, the fluctuation ranges of roll angle ϕ_t and pitch angle θ_t of the tag are about $[-0.05, 0.05]$ rad and $[-0.07, 0.05]$ rad,

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(a) Actual motion of platform



(b) Pose changes of platform

Fig. 15. Motion of the deck simulation platform.

respectively. Fig. 9(e) presents the time evolution of (u_1, v_1) of four corner points, where the black dashed lines represent the boundaries, i.e. ± 640 and ± 360. It can be observed that the lines are discontinuous at about 57 s, which is caused by the switch of detected tag. Moreover, since $\psi \approx 0$ after 50 s based on Fig. 9(c), then u_2 of point p_2 and p_3 will be similar, which results in their curves to be almost coincident, and the same to other points. Furthermore, v_1 during [10, 40] s is close to the boundary, which explains why ψ in Fig. 9(c) does not continue to decrease during the period, since this will cause the tag to be out of sight. Besides, Fig. 9(f) gives the changes of image moments in detail. For convenience of distinction, the solid lines represent the image moments at the first stage while the dotted lines correspond to the second stage, which are separated by a black dashed line at about 57 s. When reaching the black dashed line, the image moments are close to the desired value $q_d = [0, 0, 1, 0]^\top$, and the small tag has been detected. It can be seen that there exist noticeable fluctuations for q_z at the second stage, which is due to the up-and-down state of the boat caused by the motion of the waves. Moreover, since the UAV is close to the tag at this stage, the shaking of the boat will cause a large offset of the tag in the image, thereby resulting in the overshoot of $q_{x,2}$ and $q_{y,2}$.

Unlike the first case, the initial position of the landing tag in the second case lies in the upper right corner of the camera image, shown as Fig. 10(a). Moreover, other conditions remain consistent. Similarly, Fig. 10 presents the snapshots of different instants during the landing. It can also be seen that the tag is kept near the image center at about 80 s, then the accurate landing is achieved. Furthermore, Fig. 11 shows the detailed experimental results. Note that the increase of x between [25, 40] s at Fig. 11(a) is mainly caused by the convergence of ψ, shown as Fig. 11(c). According to Fig. 11(b), v_z gradually changes from -0.8 m/s to -0.4 m/s at about 60 s to achieve smooth transition between two stages. Besides, θ and ϕ in Fig. 11(c) are also less than $\theta_{\max}$ and $\phi_{\max}$, respectively. Fig. 11(d) reflects the motion of the boat, which is similar to that in the first case. According to Fig. 11(e), it is obvious that the tag is always kept within FOV during the landing. Finally, Fig. 11(f) indicates that the image moments are close to q_d at about 60 s, and then the optimization control is changed from $\mathcal{P}_1$ to $\mathcal{P}_2$. After about 20 s of adjustment, the image moments are close to q_d again, and then the motors are shut down to make the UAV fall vertically.

C. Comparative Experiments

In order to embody the superiority of our method, the comparative experiments are conducted in this part. More specifically, the experiments focus on scenarios where the tag is located near the image edge during the landing process. At first, we consider the visibility constraint directly in the virtual image plane I_v, used in [27]. Fig. 12(a) shows the snapshots in this case. At about 8 s, the tag has reached the bottom of the image. However, it still continues to move downwards and ultimately results in the loss of the tag. Note that once the tag cannot appear completely in the image, it is unable to be detected, and we call it lost.

More details are shown in Fig. 13. It can be seen that the data after 8.6 s is missing, which is caused by the loss of the tag. To account for the reason, we compare the coordinates of corner points in the virtual image plane and real image plane. According to Fig. 13(b), v_1 has reached 355 pixels at the last moment (denoted by t_l) before the tag is lost. However, its corresponding v_2 is just about 346 pixels in Fig. 13(a). As a result, the rotor UAV continues to yaw, shown as Fig. 13(c), thereby making the tag to move downwards to exceed the image. From 8.6 s to 9.6 s, the UAV will still execute the control command obtained at t_l according to Remark 5. After that, since the tag has still not yet been detected, the control input is given 0 to make the UAV static for the concern of safety, which leads to the shake of attitude shown as Fig. 13(c).

In comparison, Fig. 12(b) displays the snapshots of the experiment by using our method. Similarly, the tag is close to the bottom of the image at about 8 s. However, it remains within the FOV at 9 s, which makes the rotor UAV land on the boat finally. It can be observed from Fig. 13(e) that even though v_1 is close to the boundary from about 8 to 10 s, it does not exceed the image based on the partial enlarged view. Furthermore, combined with the change of attitude shown in Fig. 13(f), we find that ψ begins to increase at about 8 s, which avoids the tag to be out of sight.

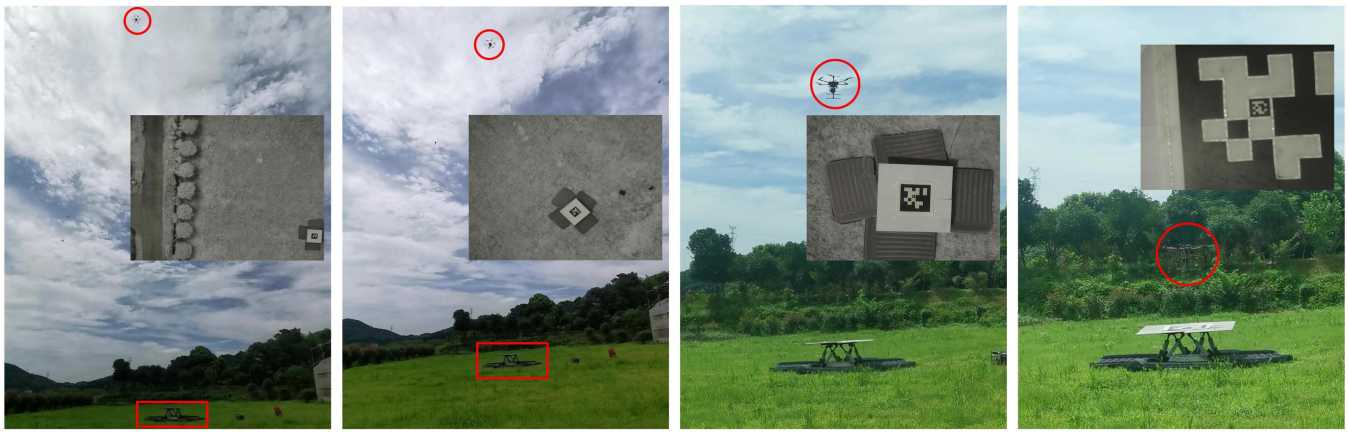
Therefore, the above experimental results indicate that compared with the method proposed in [27], our method can better keep the tag within the camera's FOV for a smooth landing.

VI. Field Experiments and Results

A. Experimental Setup

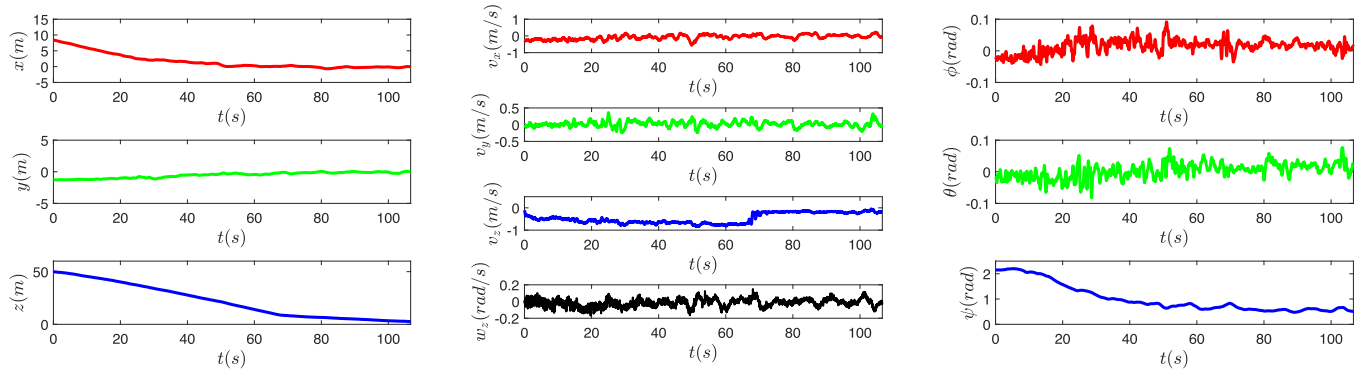
The field experimental setup is depicted in Fig. 14. The UAV used in the experiment is the DJI S900 with a wheelbase of approximately 960 mm. It has a nominal weight of 3.5 kg and

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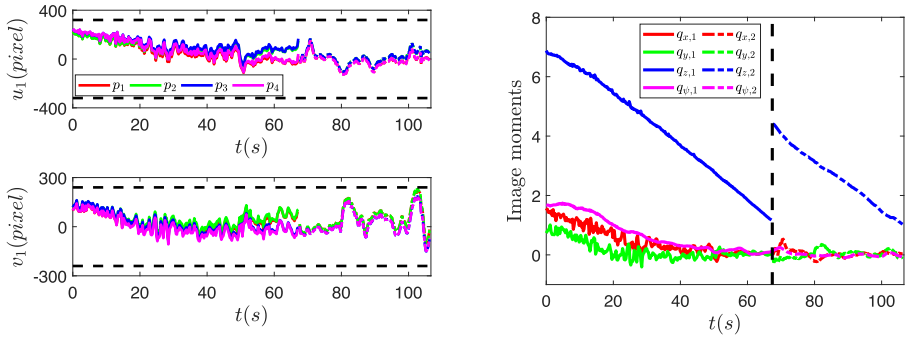


(a) t = 0 s (b) t = 35 s (c) t = 70 s (d) t = 106 s

Fig. 16. Snapshots of the landing in the first case. The red circle represents the rotor UAV, while the red rectangle represents the deck simulation platform. Besides, the picture inside is the captured image of the downward camera.



(a) Relative position (b) Velocity (c) Attitude

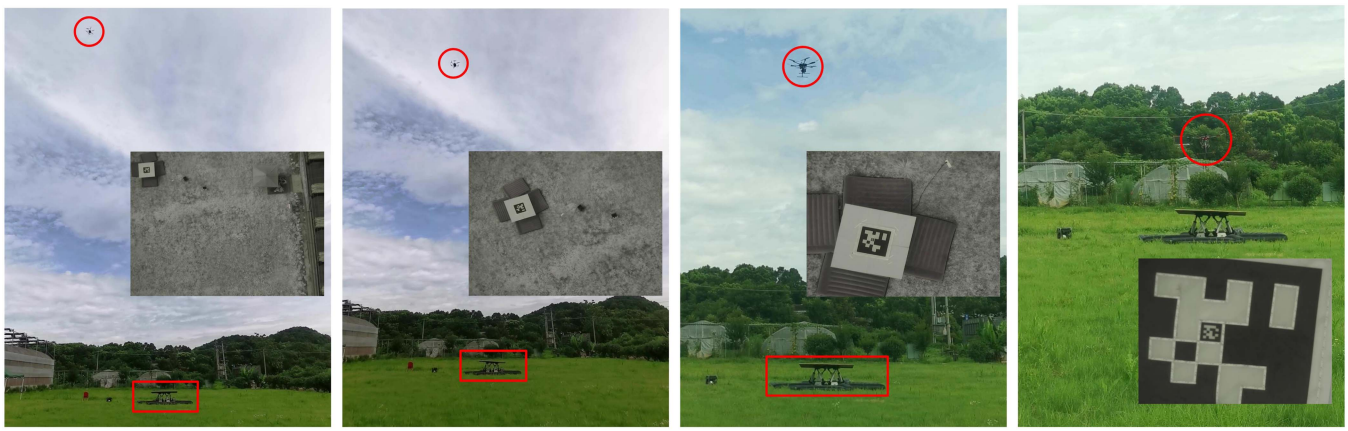


(d) (u_1, v_1) (e) Image moments

Fig. 17. Landing results corresponding to the first case.

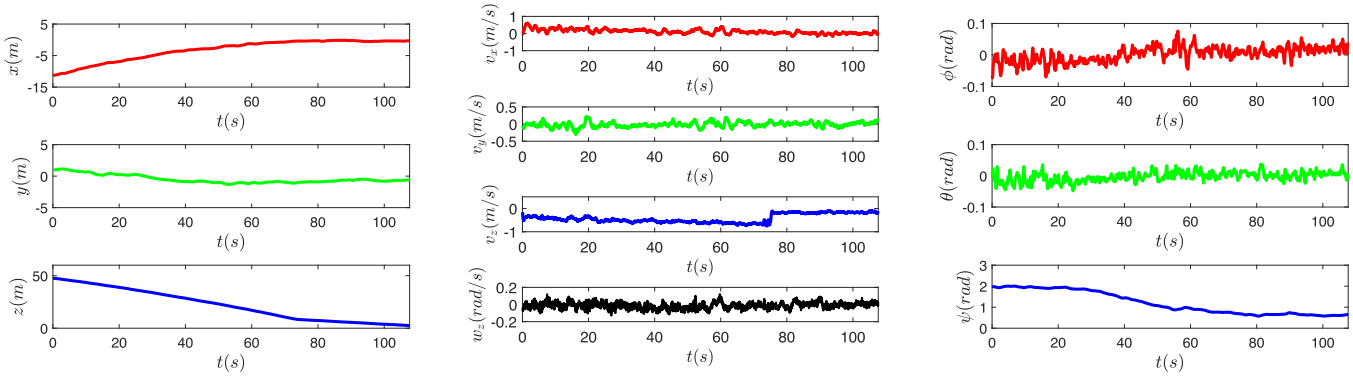
a maximum load capacity of 8.0 kg. Notably, it is equipped with a NEO v2 GPS module solely for pose data recording and landing result analysis. Moreover, the hardware includes a Nvidia Xavier NX, which integrates a 384-core Volta GPU and a 6-core Arm v8 CPU. Additionally, it incorporates an inertial measurement unit (IMU), Pixhawk flight controller and downward-looking camera. To facilitate communication, a Microhard P900 wireless modem is employed, enabling the UAV to receive commands from the custom-developed ground control station. Simultaneously, the station displays real-time UAV status information. Furthermore, DIJ Lightbridge 2 is utilized for image transmission, allowing the downward camera's

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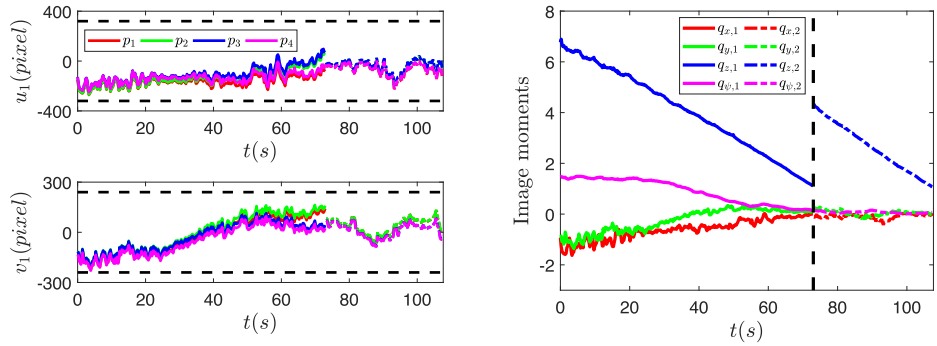


(a) t = 0 s (b) t = 35 s (c) t = 70 s (d) t = 108 s

Fig. 18. Snapshots of the landing in the second case.



(a) Relative position (b) Velocity (c) Attitude



(d) (u_1, v_1) (e) Velocity

Fig. 19. Landing results corresponding to the second case.

captured images to be visualized on a HDMI-connected screen. Considering the real-time requirements of AprilTags detection, the image is resized to 640 × 480 pixels. The program is almost directly migrated from that in the SITL simulation, with only minor changes in the interfaces and parameters, e.g. $w_u = 300$, $w_v = 220$ and l_s^* of (25) is 70 pixels in the first stage, while is 32 pixels in the second stage.

To avoid the risk of the UAV falling into the water during experiments conducted on a boat, a deck simulation platform with six degrees of freedom is employed to simulate boat motion in waves. Fig. 15 illustrates the motion of the deck simulation platform used in our experiments. Note that θ_p and ϕ_p denote the rotation angles around x_p-axis and y_p-axis in Fig. 14, respectively. The time evolution of them are shown in Fig. 15(b).

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It is apparent that both θ_p and ϕ_p exhibit a period of 10 s with an amplitude of 10°. However, their phase angles differ by $\frac{\pi}{2}$. To provide a clearer understanding of the actual motion effect of the platform, Fig. 15(a) presents the platform's state under two different inclination angles.

B. Experimental Tests

This part aims to verify the feasibility of our method in the field experiments. Two cases with different initial tag position in the image are considered. In the first case, the tag lies in the bottom right corner of the camera image, and Fig. 16 shows the snapshots of different instants. It can be seen that the tag gets close to the image center gradually. At about 106 s shown in Fig. 16(d), the UAV is close enough to the platform, and concurrently the small tag is located near the image center, then the UAV shuts down the motors to fall vertically. Furthermore, Fig. 17(a) shows the time evolution of the relative position between the UAV and the platform, which indicates that the relative error will converge to about 0. The time evolution of the UAV velocity is presented in Fig. 17(b), it is obvious that the switch of two stages is at about 67 s. Due to the effect of wind disturbance in the field experiments, there exists noticeable shaking in the pose of the UAV shown in Fig. 17(c). Despite this, our method can still guarantee the accurate landing of the UAV on the deck simulation platform. Besides, it can be seen that all the corner points are kept within the image range according to Fig. 17(d), and the large offset of v_1 after about 80 s is mainly due to the motion of the platform and the approach of UAV to the platform. More specifically, the closer the camera is to the tag, the greater the offset caused by the same motion of the tag. Even so, the UAV can still be controlled to keep the tag near the image center, thereby achieving accurate landing. Finally, the changes of image moments are shown in Fig. 17(e). It is obvious that both q_u and q_v change drastically, which are directly affected by the change of attitude shown in Fig. 17(c).

In the second case, the initial position of the tag lies in the upper left corner of the camera image, shown as Fig. 18(a). More snapshots at time instants $t = 35$ s, $t = 70$ s and $t = 108$ s are provided in Fig. 18 as well. Furthermore, Fig. 19 shows the detailed landing results in this case. It can be observed from Fig. 19(a) and (b) that the switch of the two stages happen at about 72 s. According to Fig. 19(d), the tag is always kept within the image range with the aid of our method. Simultaneously, the image moments will approach neighborhood of q_d to achieve accurate landing, shown as Fig. 19(e).

VII. Conclusion

This paper proposes a novel method for the autonomous landing of a rotor UAV from a high altitude. By addressing the limitation of classical IBVS method that overlooks constraints, a robust online MPC-IBVS method is introduced. The impact of the UAV attitude changes on visibility constraints is analyzed, thereby better ensuring the tag is within the image range. Besides, due to the altitude insensitivity of the feature point, image moments are utilized to enhance convergence rates in altitude. The proposed algorithm is proven to be robust against bounded disturbance, with an upper bound provided when the convergence domain is specified. Furthermore, a two-stage controller is designed to facilitate fast descent and accurate landing. Extensive experiments, including numerical simulations, SITL simulations and field experiments, validate the feasibility and superiority of our approach. Comparative experiments further demonstrate its ability to maintain long-term visibility of the landing tag during the landing process.

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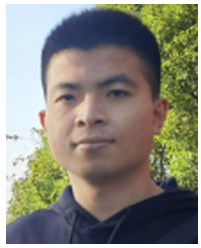
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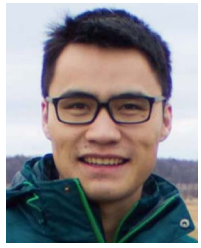
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